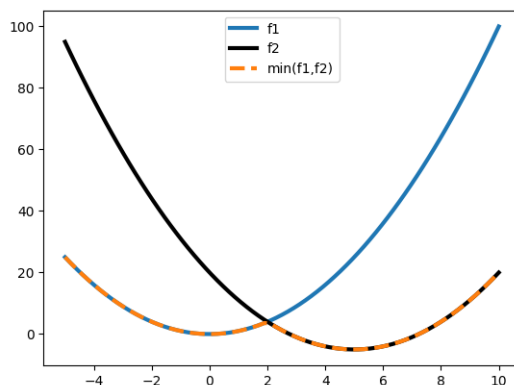


ECE 647: Lecture 10

Minimization of a jointly convex function over a convex set.

In general, a minimization of convex functions is **NOT** a convex function.



However, there is an important exception. If f is jointly convex in (x, y) and C is a convex non-empty set, then:

$$g(x) \triangleq \inf_{y \in C} [f(x, y)] \text{ is convex.}$$

Difference with maximization:

- f must be jointly convex
- Minimum must be taken over a convex set (e.g. can't just take two functions)

Proof: Assume that the infimum is always attained (What to do if this assumption does not hold?). Take:

$$g(x_1) \triangleq \inf_{y \in C} [f(x_1, y)] = f(x_1, y_1)$$

$$g(x_2) \triangleq \inf_{y \in C} [f(x_2, y)] = f(x_2, y_2)$$

We want to show that

$$g(\theta x_1 + (1 - \theta)x_2) \leq \theta g(x_1) + (1 - \theta)g(x_2)$$

$$\Longleftrightarrow$$

$$\inf_{y \in C} [f(\theta x_1 + (1 - \theta)x_2, y)] \leq \theta f(x_1, y_1) + (1 - \theta)f(x_2, y_2)$$

It suffices to find a $y \in C$ s.t.

$$f(\theta x_1 + (1 - \theta)x_2, y) \leq \theta f(x_1, y_1) + (1 - \theta)f(x_2, y_2)$$

Taking $y = \theta y_1 + (1 - \theta)y_2$, we have $y \in C$ and the result follows from the convexity of f .

Examples:

(a) The distance to a set: $\text{dist}(x, C) \triangleq \inf_{y \in C} \|x - y\|$

If C is a convex set, then $\text{dist}(x, C)$ is convex.

(b) $g(x) = \inf [f_0(y)]$ s.t. $f_1(y) \leq x$

If $f_0 + f_1$ are convex, then g is convex.

Proof: Define $f(x, y) = \begin{cases} f_0(y) & f_1(y) \leq x \\ \infty & \text{otherwise} \end{cases}$

-This is the "extended" function that we will see again in composition.

Then $f(x, y)$ is convex in (x, y) and $f(x) = \inf_{y \in \mathbb{R}} [f(x, y)]$
(prove this yourselves)

-Useful when we study strong duality.

Perspective of a function

If f is convex, then its perspective

$$g(x, t) = tf\left(\frac{x}{t}\right), \text{ dom}(g) = \left\{\frac{x}{t} \in \text{dom}(f), t > 0\right\}$$

is also convex.

Proof: Recall perspective mapping $(x, t) \rightarrow \frac{x}{t}$

$$\text{epi}(g) = \{(x, t, s) | s \geq tf\left(\frac{x}{t}\right)\} = \{(x, t, s) | \frac{s}{t} \geq f\left(\frac{x}{t}\right)\}$$

$$\text{epi}(f) = \{(x, s) | s \geq f(x)\}$$

Hence, $\text{epi}(g)$ is the inverse perspective-mapping of $\text{epi}(f)$.
 $\implies \text{epi}(g)$ is convex.

More directly:

$$\text{Let } s_1 = g(x_1, t_1) = t_1 f\left(\frac{x_1}{t_1}\right) \implies \frac{s_1}{t_1} = f\left(\frac{x_1}{t_1}\right)$$

$$s_2 = g(x_2, t_2) = t_2 f\left(\frac{x_2}{t_2}\right) \implies \frac{s_2}{t_2} = f\left(\frac{x_2}{t_2}\right)$$

Consider: $\theta s_1 + (1 - \theta)s_2, \theta x_1 + (1 - \theta)x_2, \theta t_1 + (1 - \theta)t_2$

We want to show that

$$\theta s_1 + (1 - \theta)s_2 \geq [\theta t_1 + (1 - \theta)t_2]f\left(\frac{\theta x_1 + (1 - \theta)x_2}{\theta t_1 + (1 - \theta)t_2}\right)$$

$$\Longleftrightarrow$$

$$\frac{\theta s_1 + (1 - \theta)s_2}{\theta t_1 + (1 - \theta)t_2} \geq f\left(\frac{\theta x_1 + (1 - \theta)x_2}{\theta t_1 + (1 - \theta)t_2}\right)$$

We hope to write $\frac{\theta u_1 + (1 - \theta)u_2}{\theta t_1 + (1 - \theta)t_2}$ as a convex combination of $\frac{u_1}{t_1}$ and $\frac{u_2}{t_2}$

($\frac{u_1}{u_2}$ can be either $\frac{s_1}{s_2}$ or $\frac{x_1}{x_2}$).

If we can do that, then we will just need $\gamma \frac{s_1}{t_1} + (1 - \gamma) \frac{s_2}{t_2} \geq f(\gamma \frac{x_1}{t_1} + (1 - \gamma) \frac{x_2}{t_2})$ which follows from the convexity of f .

This convex combination is:

$$\frac{\theta u_1 + (1 - \theta)u_2}{\theta t_1 + (1 - \theta)t_2} = \frac{u_1}{t_1} \frac{\theta t_1}{\theta t_1 + (1 - \theta)t_2} + \frac{u_2}{t_2} \frac{(1 - \theta)t_2}{\theta t_1 + (1 - \theta)t_2}$$

where we can substitute $\gamma = \frac{\theta t_1}{\theta t_1 + (1 - \theta)t_2}$ (note gamma is independent of u) simplifying to:

$$\frac{\theta u_1 + (1 - \theta)u_2}{\theta t_1 + (1 - \theta)t_2} = \gamma \frac{u_1}{t_1} + (1 - \gamma) \frac{u_2}{t_2}$$

so we are done.

Example:

(a) $R(p) = \log(1 + \frac{p}{N})$

- Serve the user for only $0 < t < 1$ fraction of time at average power P
- Peak power is $\frac{P}{t}$
- Avg. rate = $R(p, t) = t \cdot \log(1 + \frac{p}{t})$
-jointly concave in p, t

(b) $\frac{x^2}{t}, \frac{x^4}{t^3}, t > 0$, are both convex

$\frac{x^T x}{v}$ is convex

$\sqrt{xy}$ is concave

(c) KL-divergence $\sum_i p_i \log(\frac{p_i}{f_i})$

(d) Combined with an affine mapping, if f is convex, then:

$$g(x) = (c^T x + d) f\left(\frac{Ax+b}{c^T x + d}\right) \text{ is convex}$$

$$\text{with } \text{dom}(g) = \{c^T x + d > 0, \frac{Ax+b}{c^T x + d} \in \text{dom}(f)\}$$